Gravitational Dominance in Planetary Systems: A Force- Based Hill-Radius Scaling Approach

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Abstract

Determining the extent of orbital stability in multi-body gravitational systems remains a fundamental challenge in celestial dynamics. In this study, we analyze how gravitational dominance transitions with distance by directly comparing the local influence of a celestial body to that of the Sun. A numerical simulation framework is developed in which distances are normalized by the Hill radius, enabling a unified comparison across systems that differ by orders of magnitude in mass and scale. The analysis includes Earth, Mars, Jupiter, and the near-Earth asteroid 101955 Bennu, providing a broad range of gravitational environments. The results show that Hill-radius normalization reveals a consistent scaled pattern in stability behavior across all cases, with similar profile shapes and well-defined transition regions separating body-dominated and solar-dominated regimes. For Earth and Mars, these transitions occur at approximately 0.17 and 0.12 Hill radii, respectively, while Jupiter maintains stability out to nearly 0.45 Hill radii; in contrast, no stability boundary is detected for Bennu within the sampled range. These findings indicate that Hill-radius scaling provides a robust and physically meaningful framework for comparing gravitational environments, while deviations in low-mass regimes suggest that additional dynamical factors influence orbital stability beyond simple scaling.

1. Introduction

Orbital stability is a central problem in celestial dynamics within planetary systems (Murray & Dermott, 1999; Binney & Tremaine, 2008). The motion of particles and satellites is governed by the combined gravitational influence of multiple bodies, resulting in highly nonlinear dynamical behavior. In such environments, determining whether an object remains in a stable orbit or becomes unstable over time remains a key challenge. Understanding these stability conditions is essential for interpreting the structure and long-term evolution of planetary systems.

One of the primary concepts used to describe gravitational influence in such systems is the Hill radius. It defines the region around a body within which its gravitational influence dominates over that of a more massive central object. This provides a natural scale for distinguishing between stable and unstable motion. For this reason, the Hill radius is widely applied in studies of planetary satellites, asteroid dynamics, and planetary system formation.

However, comparing the gravitational influence of different celestial bodies using the Hill radius is not trivial, as these bodies vary significantly in mass and orbital distance from the central star. Bodies in planetary systems span a wide range of masses, from giant planets to small asteroids, leading to large differences in gravitational strength. As a result, absolute distance alone does not provide a consistent basis for comparison, since the same distance can correspond to very different dynamical conditions. Differences in mass and orbital configuration further complicate such comparisons. These factors make it difficult to identify common patterns in orbital stability when systems are considered on their original physical scales, and highlight the need for a normalized approach.

This study investigates orbital stability in the vicinity of different celestial bodies using a numerical simulation framework. Simulations were carried out for Jupiter, Earth, Mars, and the asteroid 101955 Bennu, which together represent a wide range of masses and gravitational environments within the Solar System. These bodies were selected to examine how stability behavior changes across systems with very different physical scales. To enable a consistent comparison between them, distances were expressed in units of the Hill radius, so that each system could be analyzed within a common, normalized framework. This allows stability behavior to be evaluated independently of absolute size or mass differences. By analyzing how orbital stability varies with scaled distance, this study examines whether similar patterns emerge across all cases and whether Hill-radius normalization reveals a shared structure in gravitational influence despite large differences in their physical properties.

2. Theoretical Background

2.1 The Hill Radius and Its Physical Meaning

The Hill radius defines the region around a celestial body within which its gravity dominates over that of a much more massive central object, such as a star. It provides a practical boundary for determining whether nearby particles remain bound to the body or are significantly affected by the central mass. The Hill radius is expressed as

$$R_H = a(m/3M)^{1/3}$$

where a is the orbital distance, m is the mass of the body, and M is the mass of the central star. This relation shows that the Hill radius increases with orbital distance and planetary mass, and decreases as the mass of the central object increases. The expression arises from the balance between gravitational attraction and tidal forces, a fundamental concept in celestial mechanics (Murray & Dermott, 1999; Binney & Tremaine, 2008). The Hill radius therefore serves as a characteristic scale for analyzing gravitational influence and orbital stability in planetary systems.

2.2 Gravitational Dominance and Orbital Stability

Gravitational dominance describes the extent to which a celestial body influences the motion of nearby particles in the presence of a dominant central mass. In such systems, particle motion is governed by both the gravitational attraction of the smaller body and the field of the central star. The relative importance of these influences varies with distance, so neither body exerts complete control over all regions of space. As a result, particle behavior depends on the relative strength of these competing gravitational influences (Murray & Dermott, 1999; Binney & Tremaine, 2008).

When the gravitational influence of a smaller body is sufficiently strong, particles can remain bound in stable orbits over long timescales, and in some cases form satellite systems. In these regions, motion is determined primarily by the local gravitational field rather than external perturbations. This balance varies with distance, and as distance increases, the influence of the central star becomes more pronounced. Particle trajectories then become increasingly sensitive to perturbations, so that even small differences in initial conditions may lead to significant deviations, including escape. At greater distances, the central star dominates the motion, and particles are no longer bound to the smaller body. This gradual transition in dynamical behavior provides the basis for analyzing orbital stability as a function of distance, as implemented in the present model.

2.3 Applications in Astrophysical Systems

Gravitational dominance plays a central role in a variety of astrophysical systems, including planetary satellite formation, asteroid dynamics, and protoplanetary disk evolution. In these contexts, it determines how bodies interact gravitationally and whether stable structures can form and persist. These applications illustrate the broader relevance of gravitational dominance as a framework for understanding orbital behavior across a range of dynamical environments (Rasio & Ford, 2008).

3. Methods

3.1 Computational Framework

A numerical model was constructed to evaluate the relative gravitational influence of selected celestial bodies on nearby test particles, implemented in Python using standard numerical and visualization libraries under controlled and reproducible conditions. In the simulation, a test particle of negligible mass is placed at a sequence of radial distances from each body, and at each position the gravitational contributions of both the body and the Sun are evaluated to determine which dominates locally. Orbital eccentricity and non-gravitational effects are neglected, consistent with a first-order treatment focused solely on gravitational interactions. Gravitational forces are calculated using Newton's law of universal gravitation for both body–particle and Sun–particle pairs, and by sampling across a continuous range of distances, the model resolves how relative gravitational influence varies with position. This approach provides a first-order approximation of gravitational dominance without resolving full orbital trajectories.

3.2 Stability Ratio

Orbital stability is quantified using a dimensionless stability ratio defined as:

$$S = \frac{F_{\text{body}}}{F_{\text{solar}}}$$

where F_{body} and F_{solar} are the gravitational forces exerted by the body and the Sun, respectively. Values of S > 1 indicate regions in which the body's gravitational influence exceeds that of the Sun, while S < 1 identifies regions where solar perturbations dominate. Evaluating this ratio across the sampled distances provides a direct measure of the transition between these regimes.

3.3 Hill Radius Scaling

To compare systems with substantially different masses and orbital distances, all radial positions are expressed in units of the Hill radius R_H. This normalization defines a dimensionless coordinate system in which gravitational influence can be examined independently of absolute scale.

The Hill radius is given by:

$$R_H = a\left(\frac{m}{3M}\right)^{1/3}$$

where a is the semi-major axis of the body, m its mass, and M the mass of the central star.

Sampling is performed over the interval $0.01 \leq r/R_H \leq 2$, spanning the region from strongly bound orbits to distances where solar effects become significant. This range captures the transition between body-dominated and star-dominated gravitational regimes.

3.4 Simulated Systems

The analysis includes four bodies selected to represent distinct gravitational regimes: Earth and Mars as terrestrial planets, Jupiter as a high-mass gas giant, and the asteroid 101955 Bennu as a low-mass small body.

Earth (5.97×10^{24} kg, 1 AU) has a Hill radius of approximately 0.01 AU. Mars (6.42×10^{23} kg, 1.52 AU) has a smaller Hill radius of about 0.005 AU. Jupiter (1.90×10^{27} kg, 5.2 AU) exhibits a substantially larger gravitational domain, with a Hill radius near 0.35 AU. In contrast, 101955 Bennu (7.3×10^{10} kg, 1.13 AU) has a Hill radius on the order of 1.5×10^{-6} AU, reflecting the weak gravitational field of small bodies.

For each system, distances are normalized by R_H, and the stability ratio is evaluated over the same dimensionless range, enabling direct comparison across systems that differ by many orders of magnitude in mass and size. All simulations are performed using identical numerical settings and assumptions, ensuring that differences in the resulting stability profiles reflect the physical properties of the systems rather than the computational procedure.

4. Results

The simulation results describe how the gravitational influence of each body varies with distance from its center. For all systems, this behavior is expressed in terms of the stability ratio as a function of distance normalized by the Hill radius. The following subsections present the computed Hill radii and the corresponding stability profiles for Earth, Mars, Jupiter, and the asteroid 101955 Bennu.

4.1 Hill Radius Comparison

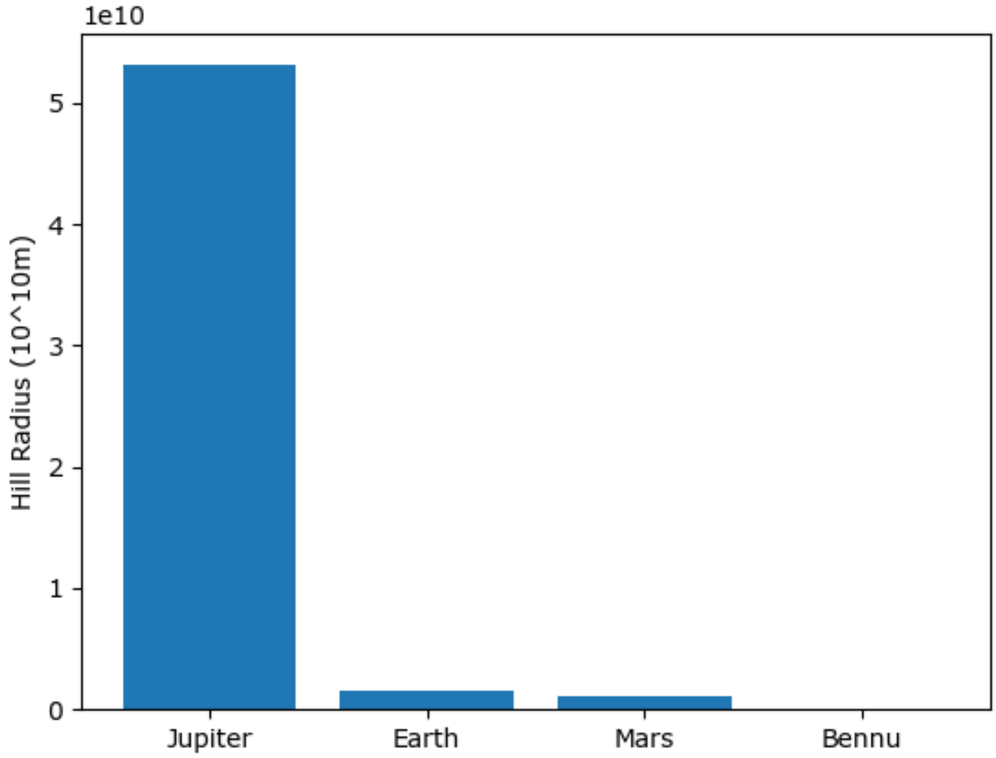

Figure 1. Comparison of Hill radii for Jupiter, Earth, Mars, and asteroid 101955 Bennu, showing the large variation in gravitational influence across bodies.

A clear hierarchy is observed, with Jupiter possessing the largest Hill radius by a substantial margin, followed by Earth and Mars, while Bennu exhibits an extremely small value. These differences span several orders of magnitude and arise from the combined dependence of the Hill radius on both mass and orbital distance, so that massive planets maintain extended regions of gravitational influence whereas small bodies affect only a very limited surrounding region. This wide separation in scale makes a normalized framework necessary, and expressing distance in units of the Hill radius allows the stability behavior of these otherwise incomparable systems to be examined on a common basis.

4.2 Orbital Stability Near Earth

Figure 2 presents the variation of the stability ratio as a function of normalized distance from Earth. At small distances, the ratio remains well above unity, indicating that Earth's gravitational influence clearly dominates the motion of the test particle. With increasing distance, the ratio declines smoothly, reflecting the increasing influence of solar gravity.

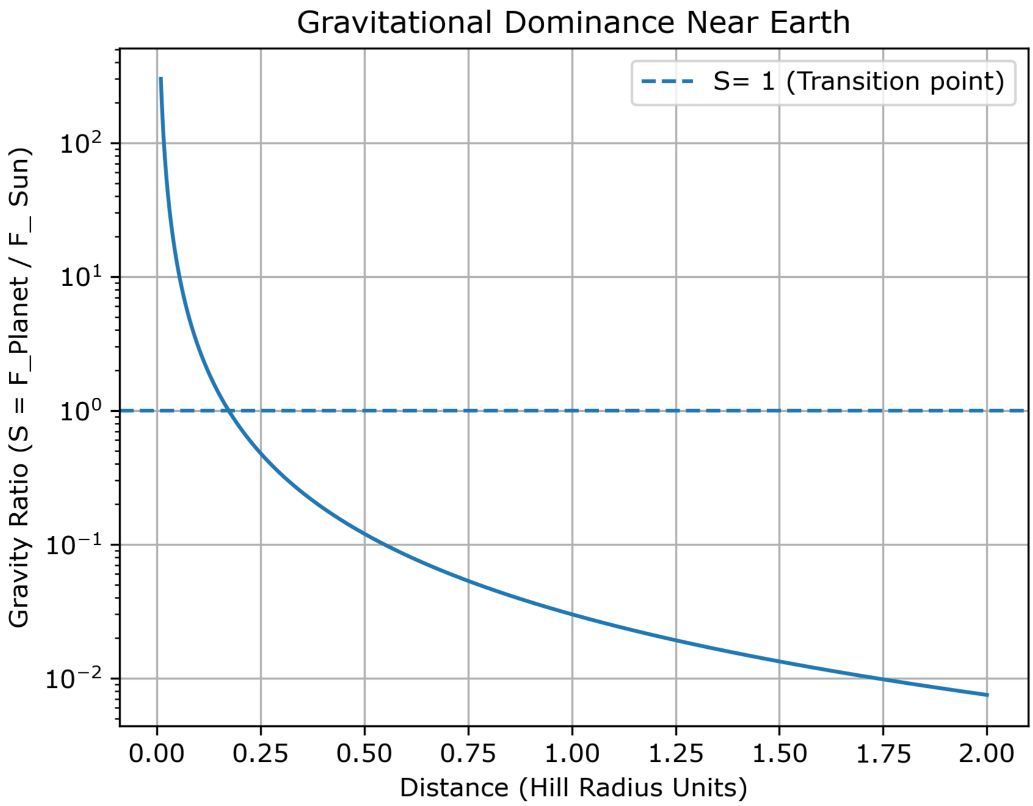

Figure 2. Stability profile for Earth as a function of normalized distance (r/R_H). The transition occurs near 0.17 R_H.

The transition point defined by $S = 1$ occurs at approximately 0.17 R_H. At this distance, the gravitational contributions of Earth and the Sun are comparable. Inside this boundary, orbital motion is primarily governed by Earth, whereas beyond it, solar perturbations become increasingly significant.

This result indicates that the region of effective gravitational dominance does not extend throughout the entire Hill sphere in this simplified model, consistent with previous studies (Domingos et al., 2006). Instead, the Hill radius acts as an upper scale, while the dynamically dominant region is more restricted.

4.3 Orbital Stability Near Mars

The stability profile for Mars, shown in Figure 3, follows a trend similar to that observed for Earth, but with a noticeable shift in the transition region. Close to the planet, the stability ratio exceeds unity, indicating local dominance of Martian gravity. As distance increases, the ratio decreases steadily, showing the increasing influence of the Sun.

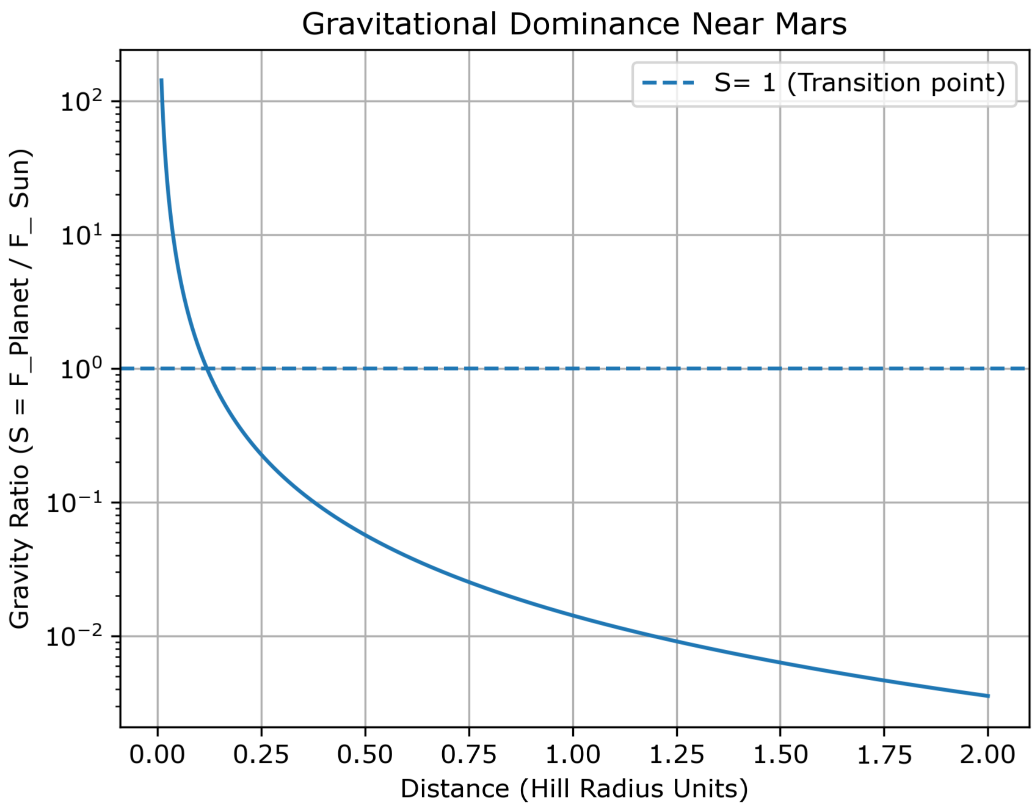

Figure 3. Stability profile for Mars. The transition point at approximately 0.12 R_H reflects a more limited region of gravitational dominance.

The equality condition $S = 1$ is reached at approximately 0.12 R_H, which is lower than the corresponding value for Earth. This inward shift reflects the weaker gravitational field of Mars, resulting in a more limited region within which it can maintain dynamical control over nearby particles. Although the overall shape is similar to that of Earth, the reduced extent of the stable region highlights the dependence of gravitational dominance on planetary mass, indicating that bodies with lower mass exhibit a more rapid transition to solar-dominated behavior when distances are expressed in normalized units.

4.4 Orbital Stability Near Jupiter

Figure 4 illustrates the stability ratio for Jupiter as a function of normalized distance. In the inner region, the ratio is substantially greater than unity, indicating strong dominance of Jovian gravity over solar influence. Although the ratio decreases with increasing distance, this decline is more gradual compared to the terrestrial cases.

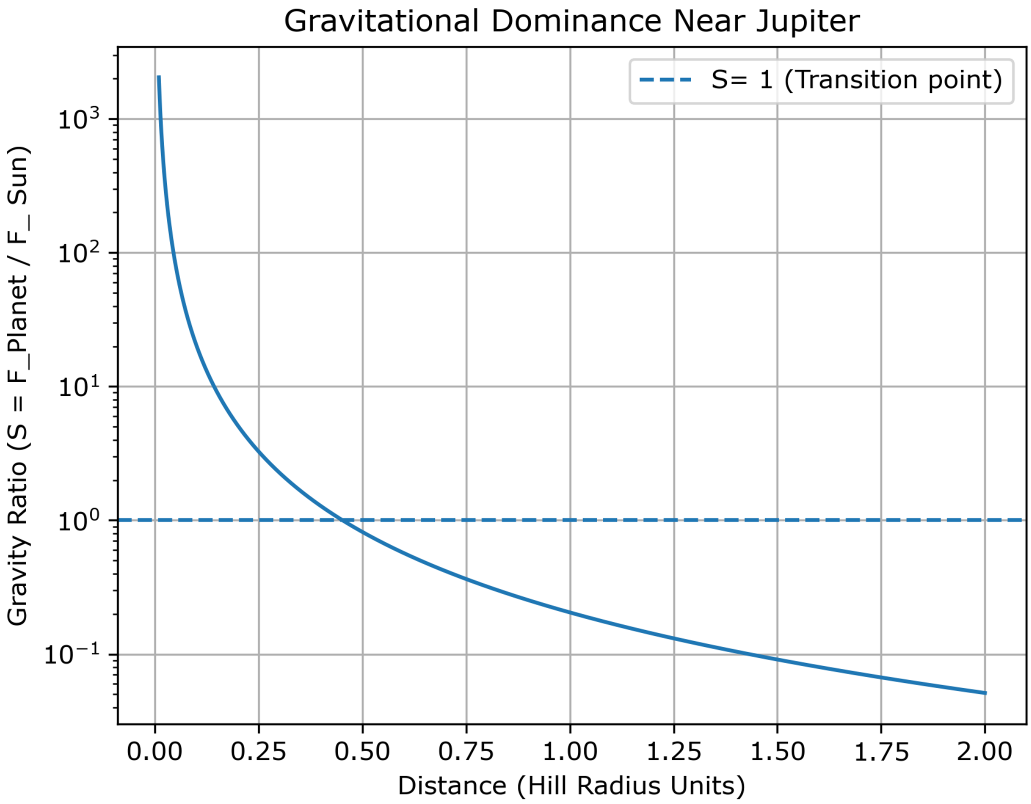

Figure 4. Stability profile for Jupiter. The transition occurs near 0.45 R_H, indicating a significantly extended region of gravitational dominance.

The transition defined by $S = 1$ occurs at approximately 0.45 R_H, significantly farther from the planet than in the cases of Earth and Mars. This outward shift indicates that Jupiter maintains gravitational control over a much larger fraction of its Hill sphere. The extended region of dominance reflects its substantially greater mass, which allows it to resist solar perturbations more effectively.

Although the overall shape is consistent with the other bodies, the slower decay and shifted transition point make the scaling effect more evident. In normalized units, Jupiter does not merely follow the same pattern—it stretches it, reinforcing the role of mass in determining how far gravitational dominance persists.

4.5 Orbital Stability Near 101955 Bennu

The stability profile for 101955 Bennu, shown in Figure 5, differs clearly from the planetary cases. Across the entire sampled range, the stability ratio remains well below unity, indicating that solar gravity dominates at all considered distances.

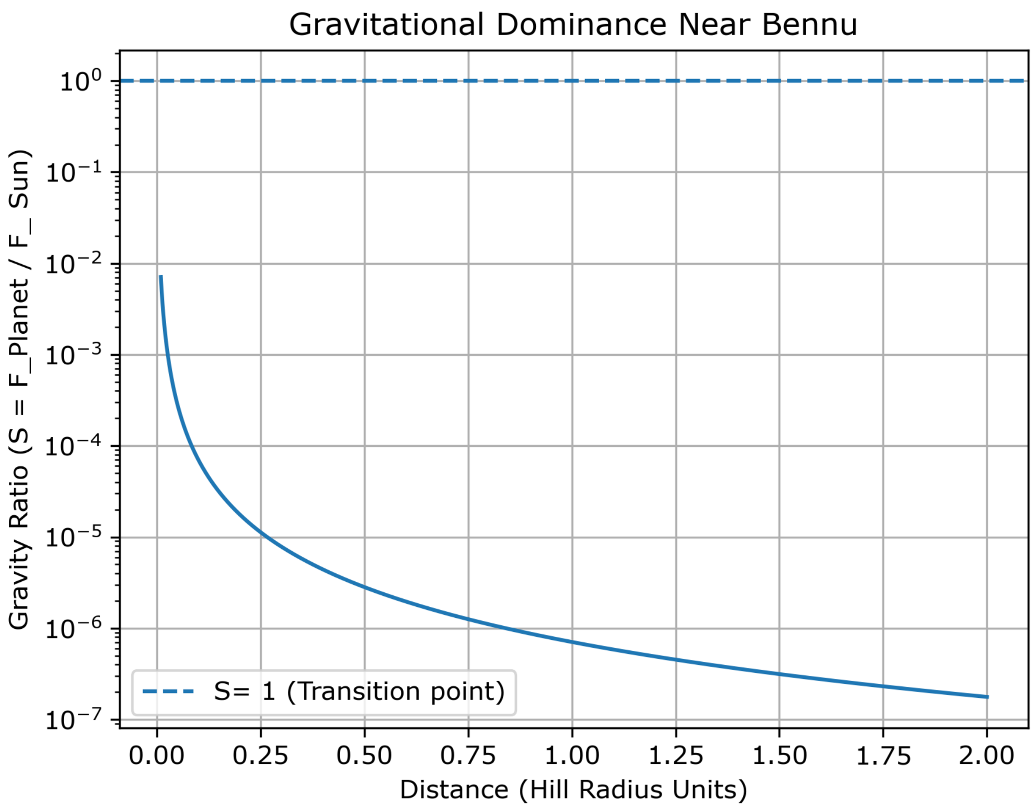

Figure 5. Stability profile for asteroid 101955 Bennu. No transition point is observed within the sampled range, indicating the absence of a dominant gravitational region.

No transition point ($S = 1$) is observed within the interval 0.01 ≤ r / R_H ≤ 2, reflecting the extremely weak gravitational influence of Bennu, which is insufficient to establish a region of dominance even at very small normalized distances. Unlike the planets, there is no distinct boundary separating body-dominated and Sun-dominated regimes. Although the resulting curve still follows a decreasing trend with distance, its position entirely below unity distinguishes it from the other cases and shows that Hill-radius scaling alone does not guarantee the existence of a stable region; for sufficiently small bodies, the normalized framework reveals a complete lack of effective gravitational control.

4.6 Comparative Behavior

Figure 6 compares the stability profiles of all simulated bodies as a function of normalized distance. When distance is expressed in units of the Hill radius, the curves exhibit a broadly similar structure despite the large differences in mass and orbital scale between the systems. In each case, the stability ratio decreases with increasing distance, indicating a consistent transition from body-dominated to solar-dominated regimes.

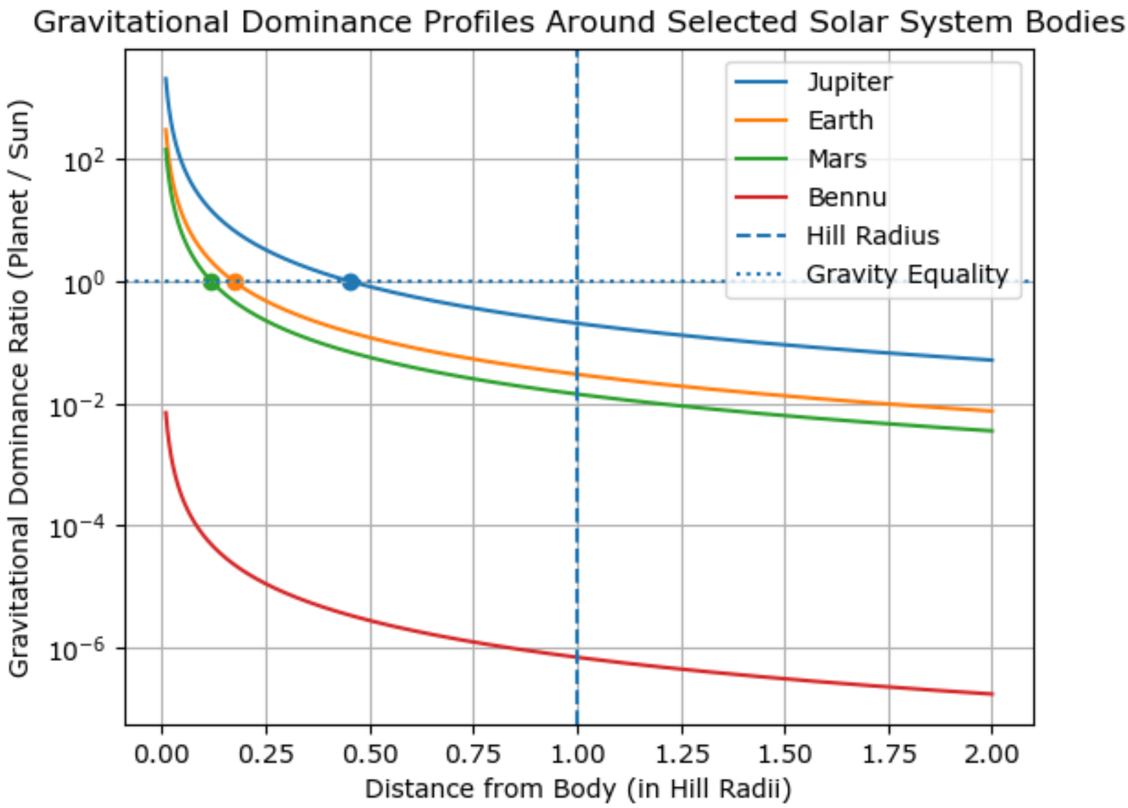

Figure 6. Comparison of stability profiles for all bodies. The curves show a similar overall structure when expressed in Hill-radius units, with deviations for low-mass bodies.

At the same time, the curves are not identical. The position of the transition point varies systematically between bodies, occurring at smaller fractions of the Hill radius for lower-mass objects and at larger fractions for more massive ones. This creates a clear ordering, with Mars transitioning closest to the body, followed by Earth, while Jupiter maintains gravitational dominance out to significantly larger normalized distances.

Bennu does not follow this pattern. Its curve remains entirely below unity across the sampled range, indicating that no region of gravitational dominance is established. This behavior separates it from the planetary cases and highlights a limiting regime in which the concept of a stable, body-dominated region becomes ineffective.

Overall, Hill-radius scaling captures the general structure of gravitational influence, but does not remove the differences between systems. Although the normalized framework reveals a shared trend, the extent of the stable region still depends strongly on mass. The Hill radius therefore serves as a useful reference scale, but not a universal boundary for orbital stability.

5. Discussion

5.1 Interpretation of Overlapping Curves

The comparison of the stability profiles shows that, when distance is expressed in units of the Hill radius, the curves for different planetary bodies follow a similar overall pattern. Despite large differences in mass and size, the stability ratio decreases with distance in a comparable way across all cases, indicating that gravitational dominance is governed primarily by general physical principles rather than the specific properties of individual bodies. This trend can be understood from the inverse-square nature of gravitational forces: as distance increases, the gravitational influence of the smaller body weakens rapidly relative to that of the Sun. The Hill radius incorporates both mass and orbital distance, effectively defining a natural scale for each system, so that when distances are normalized using this scale, the dominant physical factors are already accounted for, allowing direct comparison between different systems. As a result, the stability profiles become largely independent of absolute scale, although small deviations remain. Their overlap reflects a shared structural trend in how gravitational dominance varies with distance.

However, this agreement is not exact. Small but noticeable differences between the curves indicate that additional dynamical factors may influence stability beyond simple Hill-radius scaling. These effects become more pronounced for low-mass bodies, where the assumptions underlying the scaling are less dominant.

5.2 Implications for Orbital Analysis

Expressing distance in Hill-radius units has practical advantages for analyzing orbital stability, as it allows systems with very different physical properties to be examined within a shared framework rather than treated separately. As a result, general trends in gravitational behavior become easier to identify. In the simulations, the stability boundaries for Jupiter, Earth, and Mars occur at different fractions of their Hill radii, reflecting how mass and orbital distance affect the balance between planetary and solar gravity. At the same time, the overall similarity of the curves indicates that the same basic scaling behavior applies in each case.

This approach can also be extended beyond the Solar System, including to exoplanetary systems. By working with normalized quantities instead of absolute distances, orbital stability can be analyzed in a more general and transferable way.

5.3 Relevance for Small-Body Missions

The results for the asteroid 101955 Bennu highlight how different the gravitational environment becomes for small bodies. Unlike the planetary cases, no region is found within the sampled range where the asteroid's gravity exceeds that of the Sun. Even at distances very close to the surface, the stability ratio remains well below unity.

This behavior follows directly from the extremely low mass of Bennu and its weak gravitational field. For spacecraft operating near such bodies, solar perturbations are not a secondary effect but a dominant factor influencing motion. Maintaining stable orbits therefore requires continuous adjustment rather than relying on natural gravitational confinement.

These conditions impose strong constraints on mission design. Only a narrow range of trajectories can remain stable without active control, and even small perturbations can significantly alter orbital behavior. As a result, operations near small bodies differ fundamentally from those near planets, where extended regions of gravitational dominance exist. Even a simplified model such as the one used here illustrates why stable orbital motion is inherently more difficult to achieve in these environments.

5.4 Limitations

The analysis relies on a simplified comparison of gravitational forces and does not include full orbital dynamics. It does not solve the restricted three-body problem or perform time-dependent simulations, so it cannot capture the full complexity of satellite motion, and the stability boundaries identified here should therefore be interpreted as approximate indicators rather than exact limits. Only the gravitational interaction between the Sun and a single secondary body is considered, whereas in real systems additional perturbations from other planets can influence orbital stability, particularly over long timescales. The model also treats all bodies as spherically symmetric point masses, although small bodies such as 101955 Bennu often have irregular shapes and non-uniform mass distributions that produce more complex gravitational fields. Finally, the analysis is based on instantaneous force ratios rather than long-term trajectory evolution, whereas a more complete treatment would require numerical N-body simulations to follow particle motion over time.

Even with these simplifications, the model captures the main scaling behavior of gravitational dominance and provides a consistent framework for comparing different systems.

6. Conclusion

This study examined how gravitational dominance changes with distance from a celestial body by comparing its gravitational influence with that of the Sun. A numerical simulation framework was used, with distances expressed in units of the Hill radius to allow direct comparison between systems with very different physical scales. The results confirm that the size of stable regions depends strongly on the mass of the body, but also show that the overall structure of stability becomes more comparable when distances are normalized.

For Earth and Mars, the simulations reveal relatively sharp transition zones where the gravitational influences of the planet and the Sun become comparable. In the normalized framework, these occur at approximately 0.17 R_H and 0.12 R_H, respectively. Within these limits, motion remains largely controlled by the planet. Beyond them, solar perturbations quickly dominate, and stable motion becomes difficult to maintain. The transition is not perfectly abrupt, but the change in behavior is clear.

The case of 101955 Bennu represents a different regime. Its gravitational influence is weak across the entire sampled region, and no clear stability boundary appears within the considered range. Even at distances close to the body, solar gravity plays a dominant role. This result reflects the fundamental limitation that small bodies cannot sustain extended regions of stable gravitational control.

Jupiter shows the opposite behavior. Its large mass produces a much larger Hill sphere and a correspondingly wider region of gravitational dominance. Stable motion persists over a broad range of scaled distances, with the transition to solar-dominated behavior occurring at substantially larger fractions of the Hill radius compared to the terrestrial cases.

Taken together, these cases suggest that Hill-radius scaling captures an important part of the underlying structure of orbital stability. The curves become comparable in shape once distances are normalized, even though the physical scales differ by orders of magnitude. At the same time, the agreement is not exact. Differences remain, particularly for low-mass bodies, indicating that additional factors influence stability beyond simple scaling alone. The Hill radius therefore provides a useful and physically meaningful reference for comparing gravitational environments across a wide range of systems.

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